

Training Day 11 Report

Date: 27 June 2025

Module Name: Advanced Scanning, Enumeration & Vulnerability Analysis

Mode: Practical + Theoretical

Main Focus of the Module

Day 12 introduced advanced scanning concepts using Nmap scripts, NetBIOS/SMB enumeration commands, SNMP analysis, and vulnerability scoring systems like **CVSS**. We also discussed the **vulnerability management lifecycle**, types of scans, and tools used in the scanning and vulnerability assessment process.

Topics Covered

Nmap Scripts (NSE – Nmap Scripting Engine)

Nmap has a built-in scripting engine that allows automation of advanced scanning and vulnerability detection.

Common Script Commands Used:

```
bash
Copy code
nmap --script vuln [target]
nmap --script smb-enum-shares.nse -p 445 [target]
nmap --script snmp-info [target]
```

Example:

```
bash
Copy code
nmap -sV --script=default,vuln -T4 testphp.vulweb.com
```

Findings:

- Detected potential CVEs based on service versions
 - Enumerated SMB shares on open port 445
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Enumeration Commands

1. nmblookup – NetBIOS Name Resolution

```
bash
```

```
Copy code
nmblookup -A [target IP]
```

- Retrieves NetBIOS names and associated IPs

2. nbtstat (Windows only)

```
cmd
Copy code
nbtstat -A [target IP]
```

- Shows NetBIOS table, MAC address, user sessions

3. sgpt – ShellGPT for Intelligent Enumeration

```
bash
Copy code
sgpt "how to enumerate NetBIOS using Kali?"
```

- Used to generate and explain enumeration commands dynamically
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SNMP Check & Enumeration

SNMP (Simple Network Management Protocol): Used to monitor network devices.

- If misconfigured, it can leak sensitive info like routing tables, interfaces, etc.

Tool Used: snmp-check, snmpwalk, onesixtyone

Example Commands:

```
bash
Copy code
snmp-check -t [target IP]
snmpwalk -v2c -c public [target IP]
onesixtyone [IP] -c public
```

Findings:

- Retrieved system descriptions, OS version, and interfaces
 - Detected SNMP community string vulnerability
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Vulnerability Scoring and Lifecycle

CVSS – Common Vulnerability Scoring System

- CVSS v3.1 is an open framework for scoring severity of vulnerabilities

- Score range: 0.0 (None) to 10.0 (Critical)

Scoring Components:

- Base Score (e.g., Attack Vector, Privileges Required)
- Temporal Score
- Environmental Score

Example:

- CVE-2023-XXXX (found during Nmap scan) may have CVSS score of 9.8 → Critical

Vulnerability Management Lifecycle

1. **Discovery** – Asset inventory, scanning
2. **Assessment** – Scoring based on CVSS
3. **Prioritization** – Based on business impact
4. **Remediation** – Patching, configuration changes
5. **Verification** – Rescanning after fix
6. **Reporting** – Documentation for compliance

Types of Scanning

Type	Description
Port Scanning	Discover open ports (Nmap, Masscan)
Vulnerability Scanning	Identify known CVEs (Nessus, OpenVAS)
SNMP Scanning	Checks for misconfigurations (snmpwalk, onesixtyone)
NetBIOS/SMB Scanning	Windows-based enumeration (enum4linux, nbtstat)
Web Application Scanning	Checks web apps for issues (Nikto, Burp Suite)

Scanning Tools Used

- **Nmap** – Network scanning and scripting
 - **onesixtyone** – SNMP brute-forcer
 - **snmp-check** / **snmpwalk** – SNMP enumeration
 - **nbtstat** / **nmblookup** – NetBIOS resolution
 - **sgpt** – Real-time AI command generation
 - **OpenVAS** – (Explored theory, not used practically)
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Key Takeaways

- Advanced Nmap scripting helps identify CVEs directly
- NetBIOS and SNMP enumeration can expose valuable internal details
- Understanding CVSS scoring helps prioritize fixes
- Effective vulnerability management is a continuous lifecycle
- Always scan with proper authorization—tools are powerful and can trigger alarms